

POS Kiosk Backend — Operations Manual

What the backend service does, how to keep it healthy, and a plain-language tour of every module.

Written for

Store Owners, Managers & Non-technical Operators

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NOTE The backend has no screen of its own — it runs quietly as a Windows service and the kiosk app talks to it. This manual explains what it is doing and the few things you may occasionally need to do to it.

1. What the backend is

The “backend” is a small program that runs in the background on the same Windows machine as the kiosk. It is the brain and the filing cabinet of the POS: the kiosk screen only draws pictures and buttons — every product, price, sale, payment, refund and report is created, checked and stored by the backend.

1.1 The three pieces on the machine

Piece	Windows service name	Job
Backend service	POSBackendService	Answers the kiosk's requests; enforces the rules; generates receipts and reports.
Database	POSPostgres	Stores the data on disk (PostgreSQL 16).
Kiosk app	(not a service)	The touch screen staff use.

1.2 Where things live on disk

Folder	Contents
C:\POSKiosk\	The installed programs (kiosk app + backend\ + postgresql\).
C:\posdata\	The actual database files. This is what you back up.
C:\POSKiosk\logs\	Log files — useful for IT when something goes wrong.
C:\POSKiosk\settings.txt	One line, <code>kiosk.no=<number></code> — this lane's number, used in every invoice number.

NOTE You normally never touch the backend. It starts with Windows and keeps running. This manual covers the occasional times you do: first-day setup, checking it's healthy, backups, and restarting it.

2. First day: what is already there

A fresh install seeds a starter dataset so the kiosk can open.

2.1 The seeded administrator

Field	Value
Name (shown on the kiosk)	Cody Admin
User ID	admin
Email	admin@cody.inc
Password (API only)	password
Kiosk PIN	123456
Role	Admin

IMPORTANT Sign in on the kiosk as **CODY ADMIN** with PIN 123456, immediately set a new PIN when prompted, then create real staff accounts. On a production machine, change this account's password and PIN before going live.

2.2 Other seeded data

The installer optionally seeds sample currencies, tax categories, units of measure, and a starter set of products / modifiers. Tick **"Seed initial data"** only on the very first install; leave it unticked on upgrades so you don't get duplicates. You can replace the sample products at any time from the kiosk (Inventory → Import CSV).

3. Checking the backend is healthy

3.1 The quick check (any browser on the machine)

Open this address	A healthy system shows
<code>http://localhost:3000/api/v1/health</code>	A page of JSON with <code>"status": "ok"</code> (memory, disk and database all "up").
<code>http://localhost:3000/api/v1/health/live</code>	<code>ok</code> — the service is running.
<code>http://localhost:3000/api/v1/health/ready</code>	<code>ok</code> — the database is reachable.

If the page won't load at all, the backend service is stopped (Section 7).

3.2 The Windows Services check

- 1 Press **Win + R**, type `services.msc` , press Enter.
- 2 Find **POSPostgres** and **POSBackendService** in the list.
- 3 Both should say **Running** and **Automatic** start-up.

4. The Swagger page — looking inside the backend

The backend publishes a self-documenting web page listing every operation it can perform. You don't need it day to day, but it's the fastest way for a manager or auditor to see what the system does.

- 1 On the machine, open `http://localhost:3000/api/docs` .
- 2 You'll see a grouped list (Auth, Users, Products, Sales Orders, Payments, Reports...). Click a group to expand its operations.
- 3 Each operation has a plain-English summary, the information it needs, and an example of what it returns.
- 4 To actually try one, first get a token: expand **Auth** → **POST /auth/login/email**, click **Try it out**, enter the admin email and password, and **Execute**. Copy the `accessToken` from the response.
- 5 Click the green **Authorize** button at the top, paste the token, and close the dialog. Now **Try it out** works on the other operations.

NOTE Treat `/api/docs` as read-only unless you know what you are doing — some operations change or delete data.

5. What each module does (plain language)

The backend is organised into modules. Here is what each one is responsible for. Most of them are driven entirely from the kiosk screens described in the *Kiosk — User Manual*.

People & access

Module	Responsible for
Auth	Signing staff in (by email, user ID, or 6-digit PIN), issuing and refreshing security tokens, and the “who am I” check.
Users	The staff list: create, edit, delete, list “authorizers” (supervisors/admins who can approve voids & refunds), verify a PIN, and the public login roster the kiosk shows.
User Groups / User Permissions	Optional finer-grained permission sets that can be attached to staff.
User Details / User Addresses	Extra profile and address information for a staff member.

The menu / catalogue

Module	Responsible for
Products / Product Variants	Every sellable item and its size/price variants; CSV import of a whole menu.
Product Groups	Categories that group products, each with an optional cover image.
Modifier Groups	Add-on and choice lists (sugar level, extra shot) with min/max rules and per-option prices.
Catalog	The fast, read-only feed the kiosk uses to show the menu (active products/categories, with a separate admin feed that also shows hidden items).
Menus / Store Menus	The underlying menu structure and the per-store menu items.
Tax Categories	VAT / tax rates that can be applied to products.
Currencies	Currency definitions (symbol, code).
UOM (Units of Measure)	Units such as piece, gram, millilitre — used by inventory.

Stock & recipes

Module	Responsible for
Materials / Material Types	Raw ingredients and supplies, grouped by type.
Recipes / Recipe Items	How much of each material a product consumes when sold.
Inventory Stocks	Current on-hand quantity of each material.
Inventory Counts	Physical stock-take sessions that reconcile the system figure to a real count.

Selling

Module	Responsible for
POS Terminals	The identity of each lane (legal name, address, TIN, kiosk number) and its list of payment methods.
Sales Orders	The order itself: line items, discounts, confirm (complete the sale), void (with supervisor PIN).
Payments	Recording how an order was paid — cash (with tendered/change), card (reference + last 4), or e-wallet (reference).
Refunds	Full or partial refunds against a completed sale, with a reason and supervisor authorisation.
Discounts	The discount schemes available at checkout (e.g. Senior/PWD, promos).

Reporting

Report	Shows
Sales report (overall)	Revenue, transaction count and averages for a date range.
Hourly / Daily / Monthly	The same figures broken down by hour, day, or month.
By Product Group / By Product	Revenue and quantity per category / per item (top sellers).
By User	Sales performance per cashier.
By Payment Method	Cash vs card vs e-wallet split.
Exportable report / Mark exported	Transactions not yet sent to an external accounting system, and the action to flag them as sent.
Cashier X-Reading	A cashier's running total for today on their terminal (can be read repeatedly; can be “closed” into history).
Cashier Daily Report	The BIR-style daily summary for a cashier, closable into history.
Z-Reading	The store-wide end-of-day close, authorised, with a Z counter and sales-by-cashier.

Housekeeping

Module	Responsible for
Device Transfer	Export the entire device to one encrypted <code>.posbackup</code> file, or restore one onto a replacement machine (full replace). Admin/Supervisor only.
ERP Sync	Optional link to a back-office system: pulls the menu in and pushes orders / reports out on a schedule.
Health	The status pages in Section 3.
Cron Jobs	Scheduled background tasks — e.g. syncing ERP-managed stock levels every couple of minutes.

6. Routine jobs you may need to do

6.1 Back up the data (do this regularly)

Everything lives in `C:\posdata\`. Two ways to protect it:

1. **From the kiosk** (simplest): **Backup & Transfer** → **Export Backup** writes one encrypted `.posbackup` file. Keep it and its passphrase safe. Do this daily / weekly and before any upgrade.
2. **Database dump** (for IT): run `pg_dump` from `C:\POSKiosk\pgsql\bin\` against `pos_db` on port 5432 and copy the output off the machine.

IMPORTANT A backup file contains everyone's PINs (encrypted). Store it like a key list.

6.2 Move to a replacement machine

Use **Backup & Transfer** on both machines — export on the old one, install the software on the new one, then **Import & Restore**. Full step-by-step is in the *Kiosk — User Manual*, “Backup & Transfer”.

6.3 Free up disk space

- Old log files in `C:\POSKiosk\logs\` can be deleted when the system is healthy.
- Do **not** delete anything in `C:\posdata\`.

7. Restarting the backend

A restart fixes many transient problems and is safe — no data is lost. Try the database first if both seem stuck.

7.1 From Windows Services

- 1 Open `services.msc` .
- 2 Right-click **POSPostgres** → **Restart**. Wait until it says Running.
- 3 Right-click **POSBackendService** → **Restart**.
- 4 Re-check `http://localhost:3000/api/v1/health` .

7.2 The recovery script

If the services won't come back, run `recover-services.bat` as administrator from `C:\POSKiosk\scripts\` . It re-checks the database, re-applies any pending updates and restarts both services.

NOTE If problems persist after a restart, collect the newest lines of `C:\POSKiosk\logs\backend-error.log` and give them to your IT contact.

8. Troubleshooting

Symptom	Likely cause & action
Kiosk stuck on "Starting up"	Backend or database not running. Restart both services (Section 7); if that fails run the recovery script.
<code>/api/v1/health</code> won't load	<code>POSBackendService</code> is stopped — start it in <code>services.msc</code> .
Health page shows database "down"	<code>POSPostgres</code> is stopped or <code>C:\posdata\</code> is unreachable. Restart <code>POSPostgres</code> ; check the disk isn't full.
"Internal server error" when saving a sale	<code>C:\POSKiosk\settings.txt</code> is missing. Re-create it with <code>kiosk.no=<this lane's number></code> and restart the backend.
Invoice numbers show the wrong lane number	Wrong value in <code>settings.txt</code> . Fix it and restart the backend.
Kiosk lists no products after a fresh install	Initial data wasn't seeded. Import your menu from the kiosk (Inventory → Import CSV) or re-run the installer's seed step.
Reports are empty for a date you know had sales	Confirm the terminal is registered and that the sales were actually confirmed (not left as drafts / voided).